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PoE Power Insufficiency Causing AP to Operate in 2.4 GHz–Only Mode

Problem Summary

Several wireless access points were observed operating only on the 2.4 GHz band, with the 5 GHz radio disabled. Initial suspicion pointed to Radio Frequency (RF) configuration or controller settings, but further investigation revealed the root cause to be insufficient PoE power allocation at the access switch port. The AP's 5 GHz radio was disabled not by RF or controller settings, but by PoE under-allocation at the switch port.

Why it matters?

Modern enterprise APs dynamically disable high-power radios (5 GHz) when available PoE power is below required thresholds.

- **2.4 GHz → lower power requirement**
- **5 GHz → higher power requirement**
- **PoE under-allocation → AP enters degraded mode**

Environment

- Core Switch: Cisco Catalyst 9500 16X 10G (Backbone)
- Access Switch: Cisco Catalyst 9200L 24 PoE+
- Access Point Vendor: Hewlett-Packard Aruba AP-535 (802.11ax, 2.4 GHz / 5 GHz)
- Issue Type: PoE budget / power negotiation
- Impact: Partial wireless service degradation (5 GHz unavailable)

Lessons Learned

- **PoE is not just power delivery — it directly affects device functionality**
- **RF issues may originate from Layer 1 electrical constraints**
- **Always verify PoE budget and negotiation before changing RF or controller configurations**

Troubleshooting Process

1. **Identify the Affected Device (L3 → L2 Trace)**

```
C9500_BB# show arp | include 192.168.123.123
```

```
C9500_BB# show mac address-table address 6026.efc0.78b8
```

- Replace 192.168.123.123 with the IP address of the target device you are troubleshooting. This shows the MAC address associated with that IP.
- Replace with your actual MAC (e.g., 6026.efc0.78b8).
- This output indicates that the switch tracks MAC addresses based on the ingress interface, not the specific physical device.

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2. Locate the Access Switch via CDP

```
C9500_BB# show cdp neighbors
C9500_BB# show cdp neighbors detail
```

- Determines connected access switch and management IP

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3. Access the Edge Switch and Verify PoE Status (Critical Step)

```
P1_Incoming_01# show power inline
```

- Port showed PoE active but under-allocated
- AP powered on, but not receiving sufficient wattage

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4. Verify MAC Address on the Access Switch

```
P1_Incoming_01# show mac address-table address 6026.efc0.78b8s
```

- Confirms the exact port number where the device is connected (e.g., Gi1/0/13).

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5. Review Interface Configuration

```
P1_Incoming_01# show run interface Gi1/0/13
```

- This displays current settings for that port, including PoE, speed, duplex, and VLAN assignment.

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6. Resolution: Explicitly Allocate Adequate PoE Power

```
P1_Incoming_01# configure terminal
P1_Incoming_01(config)# interface Gi1/0/13
P1_Incoming_01(config-if)# power inline static max 30000
P1_Incoming_01(config-if)# end
P1_Incoming_01# write memory
```

- This sets the PoE budget for that port to 30 watts, ensuring sufficient power for devices such as IP cameras or access points.